

ROGRAKSHAK SYNTHETIC MEDICAL CENTER

DEPARTMENT OF PATHOLOGY & CLINICAL MICROBIOLOGY
Diagnostic Services Wing, Central Tower | Tel: +91 (22) 555-0199 | NABL & CAP Accredited Laboratory

LABORATORY MICROBIOLOGY & ANTIMICROBIAL SUSCEPTIBILITY REPORT

Patient Name:	Suresh Joshi (Downstream 1)	Specimen ID:	ACC-2026-00002
MRN:	MRN-2026-1002	Order ID:	ORD-MIC-1002
Patient ID:	PAT-0002	Specimen Type:	Sputum
Age / Sex:	49 Y / Female	Collected Date/Time:	2026-08-07 18:00
Admitting Diagnosis:	COPD Exacerbation	Received Date/Time:	2026-08-07 18:45
Report Status:	FINAL	Reported Date/Time:	2026-08-09 22:00

I. PRIMARY SPECIMEN EXAMINATION & CULTURE FINDINGS

Direct Smear / Microscopy:	Gram-negative bacilli observed (moderate to heavy). Polymorphonuclear leukocytes seen.
Culture Growth Result:	Significant Pure Growth: > 10 ⁵ CFU/mL isolated on selective and non-selective media.
Isolated Organism:	<i>Klebsiella pneumoniae</i>
Resistance Classification:	MDR
Identification Methodology:	Automated Microbial Identification (VITEK 2 GN/GP & MALDI-TOF Mass Spectrometry)

II. ANTIMICROBIAL SUSCEPTIBILITY TESTING (AST) — CLSI M100 CRITERIA

Antimicrobial Agent	MIC / Value	Interpretation	CLSI Category
Ampicillin	> 32 ug/mL	Resistant (R)	R
Amoxicillin / Clavulanate	> 32/16 ug/mL	Resistant (R)	R
Piperacillin / Tazobactam	> 128/4 ug/mL	Resistant (R)	R
Ceftriaxone	> 64 ug/mL	Resistant (R)	R
Ceftazidime	> 64 ug/mL	Resistant (R)	R
Cefepime	> 32 ug/mL	Resistant (R)	R
Meropenem	> 16 ug/mL	Resistant (R)	R
Imipenem	> 16 ug/mL	Resistant (R)	R
Ertapenem	> 8 ug/mL	Resistant (R)	R
Ciprofloxacin	> 4 ug/mL	Resistant (R)	R
Levofloxacin	> 8 ug/mL	Resistant (R)	R
Gentamicin	> 16 ug/mL	Resistant (R)	R
Amikacin	32 ug/mL	Intermediate (I)	I
Trimethoprim / Sulfamethoxazole	> 4/76 ug/mL	Resistant (R)	R
Colistin	<= 0.5 ug/mL	Susceptible (S)	S
Tigecycline	<= 0.5 ug/mL	Susceptible (S)	S

III. INTERPRETATION & CLINICAL REMARKS

*** INFECTION CONTROL ALERT: MULTIDRUG-RESISTANT ORGANISM DETECTED ***

Isolated organism *Klebsiella pneumoniae* demonstrates resistance to multiple antimicrobial classes including extended-spectrum beta-lactams, carbapenems, fluoroquinolones, and aminoglycosides.

Recommendation: Place patient under STRICT CONTACT PRECAUTIONS. Notify Hospital Infection Control Committee (HICC).
Antimicrobial stewardship consultation advised.

IV. LABORATORY AUTHORIZATION & AUTHENTICATION

Laboratory Report ID: LAB-0002	Testing Lab: Division of Clinical Microbiology	Status: FINAL
Medical Technologist: S. Nambiar, MSc (Med Micro) Reg #MLT-2021-884	Reviewing Pathologist: Dr. Arvind Deshmukh, MD (Path) Reg #MMC-2012-0498	Consultant Microbiologist: Dr. Ramesh Kulkarni, MD, DNB Verified: 2026-08-09 22:00